



LECTURE 1

WHY GEOSPATIAL REPRESENTATION LEARNING?

Geospatial Representation Learning

PRESS – OR SPACE. →

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Learning Outcomes

Lecture

- Describe the motivation and scope of geospatial representation learning.
- Explain why location, spatial context, scale, and time are central properties of geospatial data.
- Identify the main geospatial data types and applications addressed in the course.
- Recognize deep learning model families used in geospatial representation learning, including MLPs, CNNs, RNNs, transformers, and neural fields.
- Distinguish between explicit embeddings, embedding databases, location encoders, and implicit neural representations at a high level.
- Relate geospatial representation learning to environmental, socio-ecological, remote sensing, GIS, robotics, climate, agriculture, biodiversity, and urban applications.

Lab

- Load and inspect a simple geospatial dataset in Python.
- Visualize geospatial observations on a map.
- Describe how spatial location and context influence a machine learning task.
- Extract simple features from geospatial data and interpret them as first representations.
- Document the workflow and observations in a reproducible exercise report.



PRACTICAL 1

EXPLORING GEOSPATIAL REPRESENTATIONS

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